

Module 5, Quiz A (Def & Notation)

Question bank · After Read 1 · open notes · unlimited attempts

Module 5, Quiz A: Definitions & notation

Reading: [Module 5 Read 1](#)

Project: [Project 5](#) · Canvas: Def & Notation quiz (Module 5)

Work these practice items after [Read 1](#). Sections follow Read 1: scatterplot vocabulary, regression line notation (slope, intercept, prediction), residuals / MSE / R^2 , residual conditions for inference, the confidence interval for the slope, and the contrast between a confidence interval for the mean response and a prediction interval.

Full regression workflow, R output, and lab situations → [Synthesis](#).

Section A, Scatterplot vocabulary (Read 1, Build)

A1. Form

Define form in the context of describing a scatterplot.

A2. Direction

Define direction in the context of a scatterplot.

A3. Strength

Define strength in the context of a scatterplot.

A4. Pearson correlation

Define Pearson correlation r and state its range.

A5. Positive vs negative correlation

What does the sign of r tell you? What does $|r|$ close to 1 vs close to 0 tell you?

Answer key, Section A, Scatterplot vocabulary (Read 1, Build)

1. Form describes the overall shape of the pattern, whether the relationship looks roughly linear, curved, or neither (no pattern).
 2. Direction is positive (both variables tend to increase together), negative (one increases as the other decreases), or none (no consistent direction).
 3. Strength describes how closely the points cluster around a trend, strong (tight cluster), moderate, weak, or none.
 4. Pearson r is a unitless measure of the strength and direction of a linear association between two numerical variables; it ranges from -1 to 1 .
 5. $r > 0$: positive direction (as x increases, y tends to increase). $r < 0$: negative. $|r|$ near 1 : strong linear association. $|r|$ near 0 : weak to no linear association.
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Section B, Regression line notation (Read 1, Build)

B1. Fitted line notation

Write the simple linear regression prediction equation and identify each symbol.

B2. Slope interpretation

State the general interpretation template for a slope b_1 in context.

B3. Intercept interpretation

State the general interpretation template for the intercept b_0 .

B4. Point prediction

Exam-score model: $\hat{y} = 47.7 + 3.06x$ (study hours). Predict the score for a student studying 8 hours.

B5. Fitting with `lm()`

Write the R call to fit a simple linear regression of `height_cm` on `dose` using a data frame `plant_dat`.

Answer key, Section B, Regression line notation (Read 1, Build)

1. $\hat{y} = b_0 + b_1x$. $\hat{y}$: predicted (fitted) value of the response. b_0 : intercept (predicted y when $x = 0$). b_1 : slope (predicted change in y per one-unit increase in x). x : value of the explanatory variable.
 2. For each additional one-unit increase in x , the predicted value of y changes by b_1 units (increases if $b_1 > 0$, decreases if $b_1 < 0$).
 3. The intercept b_0 is the predicted value of y when $x = 0$. Caution: this is only meaningful if $x = 0$ is a plausible value in context.
 4. $\hat{y} = 47.7 + 3.06(8) = 47.7 + 24.48 = 72.18$ points.
 5. `lm(height_cm ~ dose, data = plant_dat)` (or assign to an object: `fit <- lm(height_cm ~ dose, data = plant_dat)`).
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Section C, Residuals, MSE, and R^2 (Read 1, Build and Assess)

C1. Residual definition

Define a residual for an observation (x_i, y_i) with fitted value $\hat{y}_i$.

C2. Least-squares criterion

What does the least-squares method minimize to choose the best regression line?

C3. Test MSE

Define test MSE and explain why it is preferred over training MSE for evaluating a model.

C4. R^2 interpretation

Interpret $R^2 = 0.68$ on the test set for a study-hours to exam-score model.

C5. Cross-validation

What problem does k-fold cross-validation solve, and what does it estimate?

Answer key, Section C, Residuals, MSE, and R^2 (Read 1, Build and Assess)

1. Residual $= y_i - \hat{y}_i$, the difference between the observed response and the model's predicted response.
 2. Least-squares minimizes the sum of squared residuals $\sum (y_i - \hat{y}_i)^2$, no other line produces a smaller total.
 3. Test MSE $= \frac{1}{n_{\text{test}}} \sum (y_i - \hat{y}_i)^2$ on held-out data. It is preferred because training MSE is optimistically biased, the model was fit on those same points, so training MSE understates error on new data.
 4. The linear model explains about 68% of the variability in exam scores in the test set. The remaining 32% is unexplained by study hours alone.
 5. With a small dataset a single train/test split is noisy and wastes data. K-fold cross-validation rotates every observation through the test role exactly once and averages the held-out errors, giving a more stable estimate of out-of-sample prediction error.
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Section D, Residual conditions for inference (Read 1, Assess)

D1. Normality condition

What assumption about residuals supports confidence and prediction intervals in regression?

D2. Checking residuals

Name two visual checks used to assess residual normality in regression.

D3. When inference is unreliable

If a residual plot shows a clear funnel shape (spread increases with x), what concern does this raise?

Answer key, Section D, Residual conditions for inference (Read 1, Assess)

1. Residuals should be approximately normally distributed with constant spread (homoscedasticity) across all values of x , and the observations should be independent with a linear form.
2. A histogram of residuals (should look roughly bell-shaped) and a Q-Q plot (points should fall near the diagonal line).
3. Non-constant variance (heteroscedasticity), the equal-spread condition is violated, so standard CIs and p-values may be unreliable.

Section E, Match the R output to the interpretation (Read 1, Use)

Each item shows a fitted model, one R call, and its output. Every interpretation listed is correctly worded, but only one matches the R call shown. Pick the matching letter. The trick is to recognize which quantity the call describes: the slope, the mean response at a chosen x_0 , or an individual at x_0 .

E1. `confint()` on the predictor

A model predicts exam score (points) from weekly study hours.

```
fit <- lm(exam_score ~ study_hours, data = exam_dat)
confint(fit, "study_hours", level = 0.95)
#>           2.5 %   97.5 %
#> study_hours  2.44    3.68
```

Which interpretation matches this output?

- a. Each additional hour of weekly study is associated with a change in the mean exam score of between 2.44 and 3.68 points (95% confidence).
- b. The mean exam score for all students who study 10 hours is between 2.44 and 3.68 points (95% confidence).
- c. One new student who studies 10 hours will score between 2.44 and 3.68 points (95% confidence).
- d. The mean exam score for students who study 0 hours is between 2.44 and 3.68 points (95% confidence).

E2. `predict()` with `interval = "confidence"`

Same model as E1.

```
predict(fit, newdata = data.frame(study_hours = 10),
        interval = "confidence", level = 0.95)
#>   fit   lwr   upr
#> 1 78.3  76.1  80.5
```

Which interpretation matches this output?

- a. Each additional hour of study changes the mean exam score by between 76.1 and 80.5 points (95% confidence).
- b. The mean exam score for all students who study 10 hours is between 76.1 and 80.5 points (95% confidence).
- c. One new student who studies 10 hours will score between 76.1 and 80.5 points (95% confidence).
- d. The mean exam score for students who study 0 hours is between 76.1 and 80.5 points (95% confidence).

E3. `predict()` with `interval = "prediction"`

Same model as E1.

```
predict(fit, newdata = data.frame(study_hours = 10),
        interval = "prediction", level = 0.95)
#>   fit   lwr   upr
```

```
#> 1 78.3 62.3 94.3
```

Which interpretation matches this output?

- a. One new student who studies 10 hours will score between 62.3 and 94.3 points (95% confidence).
- b. The mean exam score for all students who study 10 hours is between 62.3 and 94.3 points (95% confidence).
- c. Each additional hour of study changes the mean exam score by between 62.3 and 94.3 points (95% confidence).
- d. The mean exam score for students who study 0 hours is between 62.3 and 94.3 points (95% confidence).

Answer key, Section E, Match the R output to the interpretation (Read 1, Use)

1. a. `confint()` on the predictor row is the confidence interval for the slope, a per-one-hour change in the mean response. Option b is the mean-response CI, c is the prediction interval, and d is the intercept CI; each is worded correctly but comes from a different R call.
 2. b. `interval = "confidence"` gives the confidence interval for the mean response at the chosen $x_0 = 10$, the average score among all students who study 10 hours.
 3. a. `interval = "prediction"` gives the interval for one individual new student at $x_0 = 10$, which is why it is wider than the mean-response interval in E2.
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Section F, Choosing and using intervals (Read 1, Use)

F1. Why PI is wider

Explain in one sentence why the prediction interval is wider than the confidence interval for the mean response at the same x_0 .

F2. Which interval to use

A student wants to know what score they personally can expect to earn. Should they use the confidence interval for the mean or the prediction interval? Why?

F3. R syntax for prediction interval

Write the R call to compute a 95% prediction interval for a new observation at `dose = 7` using a fitted model `fit_plant`.

Answer key, Section F, Choosing and using intervals (Read 1, Use)

1. The prediction interval must account for both the uncertainty in the mean (same as the CI) plus the individual's random deviation from that mean, two sources of uncertainty instead of one.
 2. The prediction interval, because it covers where one individual future observation is likely to fall, not just where the average falls.
 3. `predict(fit_plant, newdata = data.frame(dose = 7), interval = "prediction", level = 0.95)`
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Section G, Testing the slope and the correlation (Read 1, Use)

G1. Slope p-value

In `summary(lm(exam_score ~ study_hours))`, the `study_hours` row shows a p-value of 0.001 for the slope. State the null hypothesis being tested and the conclusion at the 0.05 level.

G2. Same p-value

In simple linear regression, how does the p-value from the slope t-test in `summary(lm())` compare to the p-value from `cor.test()` on the same two variables? Explain why.

G3. p-value vs slope size

A model has a tiny slope estimate but a very small p-value. Does a small p-value mean the relationship is large or important? Explain.

Answer key, Section G, Testing the slope and the correlation (Read 1, Use)

1. The null hypothesis is $H_0 : \beta_1 = 0$ (no linear relationship between study hours and exam score). Since 0.001 is below 0.05, we reject H_0 and conclude there is a statistically significant linear relationship.
 2. They are identical. Testing $\beta_1 = 0$ is algebraically the same as testing $\rho = 0$, because the slope and correlation are linked by $b_1 = r(s_y/s_x)$. The two tests always give the same p-value in simple linear regression.
 3. No. A small p-value only says the slope is distinguishable from 0, not that it is large. With a big sample even a trivially small slope can be statistically significant. Judge importance from the size of the slope and the prediction quality, not the p-value alone.
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Section H, Interpreting fit on real data (Read 1, Assess)

H1. Penguins R^2

A model predicts penguin body mass (g) from flipper length (mm). On held-out test penguins, $R^2 = 0.76$. Interpret this number in context.

H2. Penguins RMSE

The same penguin model has a test RMSE of about 390 g. Interpret this number in context, and state why RMSE is often reported instead of MSE.

H3. Diamonds slope and intercept

A model predicts diamond price (\$) from carat. The fitted line is $\hat{y} = -2256 + 7756x$. Interpret the slope, and explain why the intercept is not a meaningful prediction here.

Answer key, Section H, Interpreting fit on real data (Read 1, Assess)

1. Flipper length explains about 76% of the variation in body mass among new penguins the model was not fit on. The other 24% is due to other factors and individual variation.
 2. A typical prediction for a new penguin misses its actual body mass by about 390 g. RMSE is reported because it is the square root of MSE, so it is back in the original units (grams) and easier to read than MSE in grams-squared.
 3. Slope: each additional carat is associated with a predicted increase of about \$7,756 in price. Intercept: it is the predicted price of a 0-carat diamond, which comes out negative and impossible, so it only anchors the height of the line and is not a real prediction.
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Section I, Reading conditions from plots (Read 1, Assess)

Each plot below comes from a fitted simple linear regression. Decide which condition, if any, is in trouble.

I1. Scatterplot shape

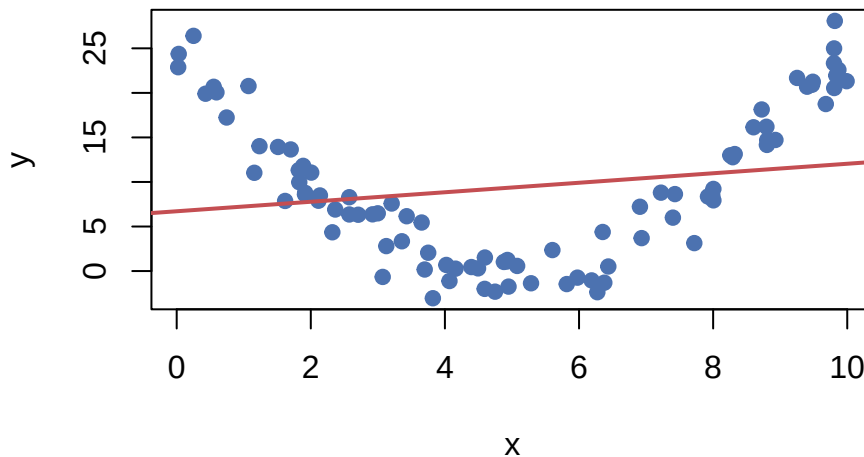


Figure 1: Data with a straight least-squares line overlaid.

The straight line is the least-squares fit. Which regression condition does this scatterplot violate, and is a slope p-value trustworthy here?

I2. Q-Q plot of residuals

The residuals curve away from the diagonal line at both ends. Which condition does this call into question?

I3. Residuals versus fitted values

The spread of the residuals grows as the fitted value grows (a funnel shape). Which condition is violated, and what does it mean for the p-value and intervals?

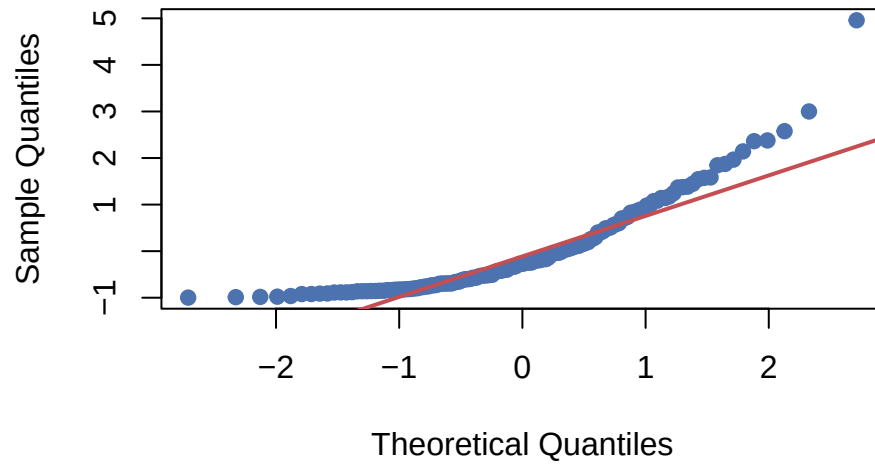


Figure 2: Normal Q-Q plot of the residuals from a fitted model.

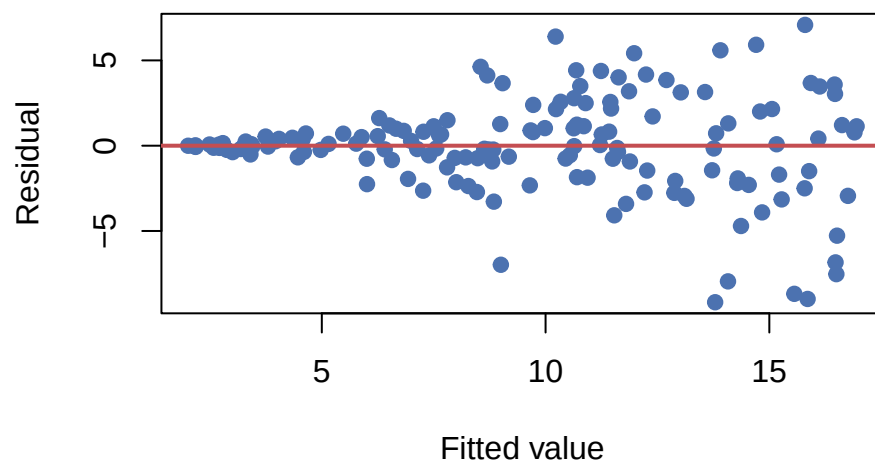


Figure 3: Residuals plotted against fitted values.

Answer key, Section I, Reading conditions from plots (Read 1, Assess)

1. The linear-form condition is violated: the true pattern is curved, not straight. A slope p-value for a line fitted to a curve is not trustworthy. Fix the form (for example, transform a variable) before doing inference.
 2. The approximate-normality condition for the residuals. The strong curvature away from the diagonal signals skewed, non-normal residuals, so confidence and prediction intervals may be unreliable.
 3. The constant-spread (equal-variance) condition is violated; this is heteroscedasticity. The standard errors are off, so the p-value and the confidence and prediction intervals are unreliable.
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